Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 2 to Draft 3

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Scope

Draft 3 (i) incorporates the two minor corrections requested in ChatGPT's referee report on the (H1) proof of Draft 2, which otherwise accepted Proposition `prop:H1`; and (ii) begins Targets B and C as formulated in Draft 0. Sections 2–11 are otherwise unchanged. The old Section 12 ("Research program") is replaced by two sections with theorems.

The two corrections

1. **Scalar uniqueness** (Corollary `cor:starK6`(b)). The commutativity-flavored argument is replaced by the referee's: from $\mathrm{Nrd}(X) = z^2\mathrm{Nrd}(v)^3$, two decompositions give $(z/z')^2 \in F^{\times 3}$, hence $z/z' \in F^{\times 3}$ because $F^\times/F^{\times 3}$ has exponent 3.

2. **Dependence on θ_0** (Corollary `cor:starK6`(c),(d); Proposition `prop:H1`). The star lift $\ell(X) = (X, \kappa(X))$ is now defined without reference to θ_0 (it is $\tilde{z}\tilde{v}^3$ with $\tilde{z} = (z, 0)$), and the formula for ψ_6 is stated for a general θ_0 with $\theta_0^3 = 1$:

$$\psi_6(X, e) = \zeta^{e-\kappa(X)}\,\theta_0(\mathrm{Nrd}\,X)^{-1},$$

   with the remark that the extra factor is a character of K^6 and so affects neither (H1) nor ℓ. This is the referee's boxed formula.

Target B (new Section 12)

Notation: $\phi_\rho = L_D\Theta_\rho$, $\Phi_\rho = \phi_\rho|_{\tilde{U}_1}$. By twisting with $\chi \circ \mathrm{Nrd}$ one normalizes $\theta_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$ according to $\rho(-1) = \pm 1$.

1. **Reduction for every row** (Theorem `thm:reduction-all`, [P]). $L_D\Theta_\rho$ is an irreducible genuine character iff Φ_ρ is a virtual character of $\tilde{U}_1$. New ingredient: the 2-group step. Every η in the preimage $\hat{P}_0$ of D_8 is good ($\eta = \eta^4(\eta^{-1})^3$), and ϕ_ρ at the lift $x_\eta = \widetilde{\eta^4}(\tilde{\eta}^{-1})^3$ equals $3\theta_0(\eta^4)\Theta_{\rho^\vee}(\eta)$. For $\rho(-1) = 1$ this is a character of D_8 (twist by $\theta_0^2 \circ \mathrm{Nrd}$). For $\rho(-1) = -1$ the relevant finite group is $\hat{P}_0$ modulo $\mu_3 \cdot \ker \mathrm{sgn}$, which is the semidihedral group SD_{16} (because $\mathrm{sgn}(-4) = \mathrm{sgn}(2) = -1$ in $\mathbb{Q}_3(\zeta_3)$), and the function $\eta \mapsto \mathrm{sgn}(\eta^4)$ is *not* a character of D_8; but it swaps the two faithful 2-dimensional characters of SD_{16}, so its product with $\Theta_{\rho'}$ (central character sgn) is a character. Checked by hand on the seven classes and by a short computation.

2. **Cube-root transform** (Lemma `lem:product`, [P]). $\phi_\rho = \phi_1 \cdot F_\rho$ on star lifts of good elements, with $F_\rho(\tilde{z}t^*) = \varepsilon_\rho(z)\Theta_\rho(t)$ on U_3, $\varepsilon_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$. This is the wild Adams operation $\psi^{1/3}$, which at a tame place is a Galois conjugation (the Draft 0 remark) and here is not a virtual character in general.

3. **Sufficient criterion** (Proposition `prop:criterion`, [P]). If F_ρ on the good elements of U_3 agrees with an $\mathrm{Ad}(U_1)$-invariant virtual character $\Psi = \sum n_i \Theta_{\sigma_i}$ of U_3, then Target B holds for ρ and $\tau_\rho|_{\tilde{U}_1} = \sum n_i \,\mathrm{Ind}_{\tilde{U}_3}^{\tilde{U}_1}(\lambda_0 \otimes \sigma_i \circ p)$. Proof: the induced character of $\lambda_0 \otimes \sigma$ is $\Theta_\sigma \cdot \Theta_{\mathrm{Ind}\lambda_0}$ by invariance, and $\Theta_{\mathrm{Ind}\lambda_0} = \Phi_1$ (Target A).

4. **All ρ trivial on K^2** (Theorem `thm:levelone`, [P^c] since it rests on Target A). For such ρ, F_ρ on good $g \in U_3$ equals $\varepsilon_\rho(z)\Theta_{\rho|K^1}(\pi_3(g))$ with π_3 the depth-three digit: the cube root of a depth-three good element has leading digit w with $w^5 = \pi_3(g)$, $w^5 = \pm w$ in $\mathbb{F}_9$, and Θ_ρ is invariant under $w \mapsto -w$ (conjugation by i). So the criterion is met by $\Psi = \varepsilon_\rho \otimes (\rho|_{K^1}) \circ \pi_3$, and

$$\tau_\rho|_{\tilde{U}_1} = \bigoplus_\psi m_\psi \,\mathrm{Ind}_{\tilde{U}_3}^{\tilde{U}_1}(\lambda_0 \otimes (\varepsilon_\rho \cdot \psi \circ \pi_3) \circ p), \qquad \rho|_{K^1} = \bigoplus m_\psi \psi,$$

   with the induced pieces irreducible and distinct (norm computation). This covers: the five D_8-representations (Corollary `cor:levelzero`), the level-zero representations with $\rho(-1) = -1$ which that corollary missed, the four 4-dimensional representations of $D^\times/F^\times K^2 \simeq \mathbb{F}_9 \rtimes D_8$ ($\dim \tau_\rho = 108$), and all their norm twists. Draft 1–2 described the level-one family as "dimension 9 or 18"; that was wrong (the group has order 72).

5. **Level ≥ 2** ([O]). Reduced to a finite $\mathbb{Z}$-linear problem per ρ on the good classes of Q_3/Q_{3m+1} (order 3^6, class two, for $m = 2$); on Q_6/Q_7 the criterion is automatic by the linearity of the inverse Artin–Schreier map, and the difficulty sits on $\bar{G}_3$ where the cube root is not linear. Not carried out.

Target C (new Section 13)

Setting: odd prime p, $F \supset \mu_p$, $e = e(F/\mathbb{Q}_p)$, $d = pe/(p-1)$, $k_0 = 2e/(p-1)$, $n = p$, E_0 unramified, cocycle twist $c \equiv -1/4$ (the scalar-centrality condition; $c = 2$ for $p = 3$).

1. **Depth facts** (Lemma `lem:depth-gen`, [P]): p-th powers $K^k \to K^{k+2e}$ bijective for $k > k_0$; $w^p \in F \Rightarrow w \in F$; $K^{2d} = U_F^{(1)} \cdot (K^{k_0})^p$ with $z \equiv \mathrm{Nrd}(X)^{(p+1)/2}$ (Artin–Schreier argument on the boundary level, uniform in q).

2. **p-th power identity** (Theorem `thm:pthpower`, [P]): $(w, 0)^p = (w^p, 0)$ for $w \in K^1$. With S_k the Lucas-type sequence $S_{k+1} = TS_k - nS_{k-1}$ (so $w^k = S_k w - nS_{k-1}$), the exponent is $\sum_{k=1}^{p-1}(S_{k+1}, S_k) + \sum_{k=2}^{p}(S_k, n)$; the Cassini identity $S_{k+1}S_{k-1} - S_k^2 = -n^{k-1}$ and Steinberg give $2(a_k - a_{k-1}) = (k-1)\Delta^2 b_k$, and summation by parts kills the sum modulo p. The combinatorial step was machine-checked for $p \leq 13$ on random data; the symbol computations themselves were not machine-checked for $p > 3$ (no p-th power symbol implementation).

3. **(H1) for every odd p** (Theorem `thm:H1-gen`, [P]): the canonical star section is a homomorphism on $F^\times K^{2d}$, with $\kappa(X) = -\frac{p+1}{2}(y_X, \mathrm{Nrd}X)_E$. The proof is the Draft 2 proof with $U^{(3)}$ replaced by $U^{(d)}$ and $m = -\frac{p+1}{2}$; every case closes because $2m + 1 \equiv 0$.

4. **Not uniform** (Remark `rem:uniform`): the Brauer reduction is expected to go through but is not written ([O]); the finite design (Section 9) is genuinely specific to (p, e, q) ([C]); composite n not examined.

Status changes

Item	Draft 2	Draft 3
Proposition `prop:H1`	[P], two minor corrections requested	[P], corrected
Reduction theorem, every row	[O] (claimed "verbatim")	[P]
Criterion for a row	—	[P]
Target B, ρ trivial on K^2 (all central characters)	partly [P^c]	[P^c]
Target B, level ≥ 2	[O]	[O], finite problem sta
p-th power identity, general p	—	[P]
(H1), general p	—	[P]
Target C, Brauer reduction / design	[C]	[O] / [C]

Items to referee with particular care

(1) The SD_{16} step in Theorem `thm:reduction-all`: the identification of the group (the values $\mathrm{sgn}(-4) = \mathrm{sgn}(2) = -1$) and the swap $f\chi_1 = \chi_2$. (2) In Theorem `thm:levelone`, the identification $w = \pm\pi_3(g)$ via Lemma `lem:cube`(c). (3) The summation by parts in Theorem `thm:pthpower`, and the claim that the Kubota formula with the p-th power symbol is the Kazhdan–Patterson cocycle for general p. (4) Lemma `lem:depth-gen`(c) for ramified tori is only sketched.

Editorial

Abstract, Section 1 and the status table updated; Section 11 unchanged except cross-references; bibliography unchanged. No new scripts; the two small checks (D_8 on $\mathbb{F}_9$, the SD_{16} swap, the summation identity) were run inline and are described in the text. American English spelling throughout.